Fair Scores via Joint Calibration: A Gated Many-Facet Rasch Model for Disentangling Judge Bias and Contamination in LLM Evaluation

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Abstract

LLM-as-a-judge evaluation corrects for judge bias and benchmark contamination independently, but sequential correction produces biased model ability estimates. We formalize this failure as an omitted variable bias (OVB) problem: when judge severity and contamination effects are both present, any estimator that models one while omitting the other yields asymptotically biased ability rankings. We propose the Gated Many-Facet Rasch Model (G-MFRM), which jointly calibrates model ability, item difficulty, judge severity, and contamination susceptibility through staged estimation with externally anchored per-item exposure indicators. Formal identifiability analysis proves that fixing exposure externally is both necessary and sufficient for parameter separation under standard multi-judge designs. Error propagation analysis establishes a critical noise threshold $\sigma_{\text{crit}} = \sqrt{\text{Var}(e_i)}$ below which ability estimates degrade by less than 6%. In synthetic experiments with 50 models, 100 items, and 8 judges, G-MFRM reduces ability RMSE by 43.8% relative to sequential estimation (Wilcoxon $p < 10^{-6}$) and achieves Kendall $\tau = 0.922$ with ground-truth rankings versus $\tau = 0.859$ for the best sequential baseline.

1 Introduction

Large Language Model (LLM) evaluation increasingly relies on LLM-as-a-judge protocols, where one or more LLM judges score model responses on structured benchmarks (Zheng et al., 2023). Two well-documented sources of distortion affect these evaluations: *judge bias*, where different judges assign systematically different scores to the same response, and *benchmark contamination*, where models achieve inflated scores on items encountered during training. Both problems have received sustained research attention, producing separate toolkits for bias calibration and contamination detection. These toolkits treat the two distortions as independent phenomena, a separation that reflects their origins in different research communities rather than any principled argument for independence. Accurate ability ranking is essential for model selection, safety evaluation, and benchmark design; systematic distortion in estimated abilities undermines all three. Yet a practitioner who needs both corrections must apply them sequentially: first correct one source of distortion, then the other. This sequential approach produces unreliable ability estimates. In synthetic experiments with 50 models, 100 items, and 8 judges, sequential estimation yields ability root mean squared error (RMSE) of 0.221 ± 0.038, while joint estimation achieves 0.124 ± 0.016, a 43.8% reduction (Wilcoxon $p < 10^{-6}$). The gap does not arise from estimation noise or algorithmic limitations of individual methods; it reflects structural confounding between judge bias and contamination that no sequential pipeline can resolve.

Existing judge bias correction methods remove systematic differences across judges without modeling contamination. CyclicJudge (Zhu et al., 2026) uses round-robin pairing to cancel average bias through balanced exposure of model pairs to all judges, but treats all residual score variation as genuine ability signal. UDA (Zhang et al., 2026) learns a neural K-factor adjustment that reduces inter-judge standard deviation by 63.4%, yet operates as a black-box transformation with no mechanism for contamination effects. On the contamination side, ConStat (Dekoninck et al., 2024) compares target model performance against a reference model to estimate item-level contamination ($r^2 = 0.94$ with true contamination levels in controlled settings), DCR (Xu et al., 2025) applies fuzzy inference to adjust scores on suspected leaked items, and LogProber (Yax et al., 2024) probes token-level log-probabilities to detect memorization signatures. These two families of methods operate in separate silos. A bias-first pipeline absorbs contamination-

induced score inflation into the ability estimates, because the bias model attributes all residual systematic variation to genuine ability differences. A contamination-first pipeline misattributes judge severity differences as contamination signals, because score elevation from a lenient judge is indistinguishable from score elevation due to memorization without a joint model. In either ordering, the first correction stage produces estimates already distorted by the unmodeled facet, and the second stage inherits this distortion with no mechanism for self-correction.

We formalize this confounding as an omitted variable bias (OVB) problem (Angrist and Pischke, 2009). Within the Many-Facet Rasch Model (MFRM) framework (Linacre, 1989), the complete data-generating process includes both judge severity d_j and a contamination interaction $c_n \cdot e_i$. Fitting a specification that includes one facet but omits the other produces asymptotically biased parameter estimates whenever the two facets are non-orthogonal. This non-orthogonality holds in practice: lenient judges (low severity d_j) amplify score inflation on exposed items, while strict judges compress it, creating systematic dependence between judge effects and contamination effects. The OVB framework predicts that the bias magnitude scales with the contamination susceptibility c_n of each model, so the most contaminated models receive the largest upward distortion in estimated ability. For practical benchmark evaluations, this means that the more a model's training data overlaps with the benchmark, the more its ability is overestimated, precisely the failure mode that contamination correction is designed to prevent. Synthetic validation confirms the predicted pattern: the Pearson correlation between sequential estimation bias and true c_n reaches $r = 0.86$ ($p < 0.001$), which shows that contaminated models are systematically overestimated under both sequential orderings (Section 2).

We propose the Gated Many-Facet Rasch Model (G-MFRM), which combines the multi-facet structure of MFRM (Linacre, 1989; Jiao et al., 2025) with the gating mechanism of Doubly-Gated Item Response Theory (DG-IRT) (Shu et al., 2013) to jointly estimate model ability, judge severity, and contamination susceptibility in a single measurement framework. G-MFRM extends the MFRM linear predictor with a multiplicative contamination gating term $c_n \cdot e_i$, where $c_n \geq 0$ captures model-specific contamination susceptibility and $e_i \in [0, 1]$ quantifies per-item exposure. When $c_n = 0$ or $e_i = 0$, the gating term vanishes and the model reduces to standard MFRM, so uncontaminated model-item pairs incur no distortion. Because the bilinear product $c_n \cdot e_i$ cannot be uniquely decomposed from response data alone, G-MFRM employs a three-stage calibration procedure: (1) standard MFRM warm-start to obtain initial parameter estimates, (2) external per-item exposure estimation from evidence independent of the response data, and (3) joint Markov chain Monte Carlo (MCMC) sampling with e_i fixed from Stage 2. We provide a formal identifiability analysis (Propositions 1–2 and Theorem 1) that proves fixing e_i externally is both necessary and sufficient for parameter separation under standard multi-judge designs (Section 3). A complementary error propagation analysis bounds the impact of Stage 2 measurement noise: when exposure estimation error satisfies $\sigma \leq 0.2$, ability RMSE degradation remains below 6%. Figure **??** provides an overview of the complete framework.

Our contributions are as follows:

1. We formalize the confounding between judge bias and benchmark contamination in LLM evaluation as an omitted variable bias problem, and show that sequential correction produces directional bias proportional to model contamination levels (Section 2, Figure 1).

2. We propose the G-MFRM framework with formal identifiability guarantees (Propositions 1–2 and Theorem 1) and quantified error propagation bounds for the staged estimation procedure (Section 3, Figure **??**, Table **??**).

3. Synthetic validation confirms that joint estimation reduces ability RMSE by 43.8% relative to sequential estimation ($p < 10^{-6}$), with Kendall rank correlation $\tau = 0.922$ versus $\tau = 0.859$ (Section 4).

4. Empirical validation on real-data evaluation with 9 LLM judges and 9 target models on Arena-Hard, comparing G-MFRM fair scores against human preference rankings from Chatbot Arena (Section 4).

2 Confounding in LLM Evaluation

The Many-Facet Rasch Model (MFRM) extends classical Rasch measurement to settings with multiple rater facets (Linacre, 1989, 1994). Originating in educational measurement, the model treats rater

variation as a nuisance facet to be estimated and removed rather than a source of noise to be averaged over. In the LLM evaluation context, where J judges score N models across I items, the MFRM specifies

$$\log \frac{P(Y_{nij} \geq k \mid \theta_n, b_i, d_j, \tau_k)}{P(Y_{nij} < k \mid \theta_n, b_i, d_j, \tau_k)} = \theta_n - b_i - d_j - \tau_k, \quad (1)$$

where Y_{nij} is the ordinal rating by judge j for model n on item i, θ_n is the ability of model n ($n = 1, \ldots, N$), b_i is the difficulty of item i ($i = 1, \ldots, I$), d_j is the severity of judge j ($j = 1, \ldots, J$), and τ_k is the threshold for category k. All facets reside on a common logit scale, so model ability is directly comparable after adjustment for item difficulty and judge severity. Jiao et al. (2025) applied this framework to compare ten LLM raters with human experts, showing that MFRM separates genuine ability from systematic judge effects in multi-judge panels.

However, standard MFRM assumes that all systematic sources of score variation are captured by the modeled facets. This assumption breaks down when some models have been exposed to benchmark items during training (Xu et al., 2025). Such contamination inflates scores on exposed items through memorization rather than genuine ability. We represent this effect as an interaction $c_n \cdot e_i$, where $c_n \geq 0$ is the contamination susceptibility of model n and $e_i \in [0, 1]$ is the exposure level of item i. The complete data-generating process becomes

$$\eta_{nijk} = \theta_n - b_i - d_j - \tau_k + c_n \cdot e_i. \quad (2)$$

When a practitioner fits the standard MFRM (Eq. 1) to data generated by Eq. 2, the contamination term is absent from the estimated model. Its influence does not vanish but is absorbed into the estimated parameters $\tilde{\theta}_n$, $\tilde{b}_i$, and $\tilde{d}_j$, which distorts all downstream inferences.

The consequence of this omission can be stated precisely.

Observation 1 (Omitted Variable Bias in MFRM)**.** *Let the full model be $\eta_{nijk} = \theta_n - b_i - d_j - \tau_k + c_n \cdot e_i$ and the short model omit the contamination term: $\tilde{\eta}_{nijk} = \tilde{\theta}_n - \tilde{b}_i - \tilde{d}_j - \tau_k$. By the omitted variable bias (OVB) formula (Angrist and Pischke, 2009),*

$$\operatorname{plim}(\tilde{\theta}_n - \theta_n) = \gamma_n \cdot c_n, \quad (3)$$

where γ_n depends on the projection of the exposure vector (e_i) onto the design matrix of the short model. The bias is non-zero whenever $\operatorname{Cov}(d_j,\, c_n \cdot e_i) \neq 0$.

Observation 1 is a direct application of the classical OVB result from Angrist and Pischke (2009) to the MFRM setting. The condition for bias, non-orthogonality between judge severity and contamination effects, holds in practice: lenient judges (low d_j) tend to amplify score inflation on exposed items, while strict judges compress it. The bias magnitude scales with c_n, so heavily contaminated models receive the largest upward distortion in estimated ability. In the linear analogue, this reduces to the short-regression bias $\tilde{\beta} - \beta = \delta \cdot \gamma$, where δ captures the auxiliary regression of the omitted variable on the included regressors. When both c_n and d_j are present, any estimate of model ability from a specification that includes one facet but omits the other carries this asymptotic bias. The Observation thus establishes not only the existence of bias but also its direction and relative magnitude across models.

This confounding undermines both sequential estimation strategies. A bias-first approach, as in CyclicJudge (Zhu et al., 2026), calibrates d_j before estimating ability, but the resulting $\hat{\theta}$ is already inflated for contaminated models because the omitted $c_n \cdot e_i$ term has been absorbed into the ability estimates. A contamination-first approach, as in DCR (Xu et al., 2025), adjusts for contamination before addressing judge effects, but judge severity is then misattributed as contamination signal. Synthetic validation confirms this failure: with $N = 50$ models, $I = 100$ items, $J = 8$ judges, $\rho = 0.6$, and 20 random seeds, the Pearson correlation between sequential estimation bias and true c_n reaches $r = 0.86$ ($p < 0.001$). As illustrated in Figure 1, the bias is not random but directional: contaminated models are systematically overestimated under both sequential orderings.

3 The G-MFRM Framework

Section 2 established that omitting contamination from the measurement model produces directional bias in ability estimates (Observation 1). We now introduce the Gated Many-Facet Rasch Model (G-MFRM), which jointly estimates ability, judge severity, and contamination susceptibility within a single measurement framework. The key design choices, staged estimation with externally fixed exposure and a regularized horseshoe prior for the contamination term, are motivated by the identifi-

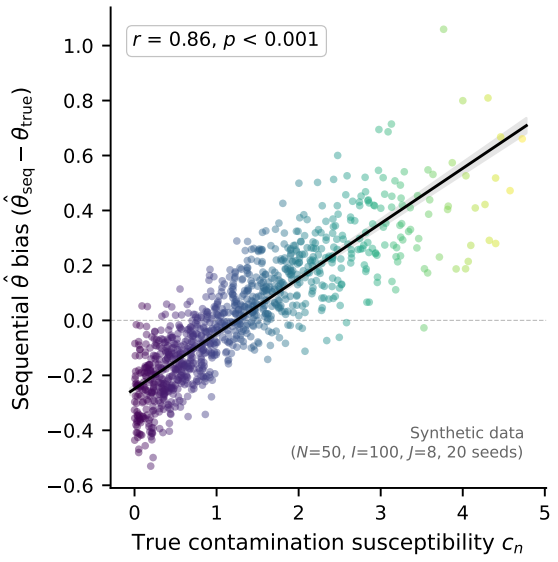


Figure 1: Sequential estimation bias ($\hat{\theta}_{\text{seq}} - \theta_{\text{true}}$) versus true contamination susceptibility c_n. Pearson $r = 0.86$ ($p < 0.001$) confirms systematic overestimation of contaminated models. Synthetic data ($N = 50$, $I = 100$, $J = 8$, 20 seeds).

ability analysis in §3.3 and the error propagation bounds in §3.4.

3.1 Model Specification

G-MFRM extends the standard MFRM (Linacre, 1989, 1994) by adding a multiplicative contamination gating term to the linear predictor (Eq. 1):

$$\log \frac{P(Y_{nij} \geq k)}{P(Y_{nij} < k)} = \theta_n - b_i - d_j - \tau_k + c_n \cdot e_i, \quad (4)$$

where all terms carry the definitions from Section 2. The gating structure $c_n \cdot e_i$ separates the model-specific and item-specific components of contamination: $c_n \geq 0$ captures how much model n benefits from memorized benchmark content, while $e_i \in [0, 1]$ quantifies how much of item i has leaked into training corpora. When $c_n = 0$ (a clean model) or $e_i = 0$ (a novel item), the gating term vanishes and Eq. 4 reduces to the standard MFRM. This graceful degradation ensures that G-MFRM imposes no distortion on uncontaminated model-item pairs.

We enforce non-negativity and sparsity of c_n through a regularized horseshoe prior (Piironen and Vehtari, 2017). A global shrinkage scale $\tau_c \sim \text{HalfCauchy}(1.0)$ controls the overall contamination level, per-model local scales $\lambda_n \sim \text{HalfCauchy}(1.0)$ allow adaptive shrinkage, and a slab component $c^2_{\text{slab}} \sim \text{InverseGamma}(2.0, 8.0)$ prevents unbounded estimates. The effective contamination is $c_n = |\tau_c \cdot \tilde{\lambda}_n \cdot c_n^{\text{raw}}|$, where $\tilde{\lambda}_n$ is the regularized local scale. This prior strongly shrinks c_n toward zero for clean models while permitting large values for genuinely contaminated ones, reflecting the expectation that contamination is sparse across the model population. The remaining priors are $\theta_n \sim \mathcal{N}(0, 1)$, $b_i \sim \mathcal{N}(0, 1)$, $d_j \sim \mathcal{N}(0, 0.5)$, and standard uniform on thresholds τ_k.

The closest antecedent is the Doubly-Gated IRT model of Shu et al. (2013), which introduced a binary gating mechanism to handle item preknowledge in educational testing. G-MFRM differs in three respects. First, exposure e_i is continuous rather than binary, allowing partial contamination to be modeled. Second, the multi-facet structure incorporates judge severity d_j as an explicit nuisance parameter, whereas DG-IRT assumes a single-rater design. Third, exposure is externally estimated rather than treated as known, necessitating the staged procedure and error propagation analysis that follow.

3.2 Staged Estimation

Joint estimation of all parameters in Eq. 4, including e_i, would require identifying the bilinear product $c_n \cdot e_i$ from response data alone. This creates a fundamental non-identifiability (Proposition 2). We resolve this through a three-stage procedure that fixes exposure externally before estimating the contamination parameters.

Stage 1: Standard MFRM. We fit the standard MFRM (Eq. 1) via MCMC (NUTS) to obtain initial estimates $(\hat{\theta}, \hat{b}, \hat{d})$. These serve as warm-start values for the sampler in Stage 3 and improve convergence.

Stage 2: Exposure estimation. Per-item exposure e_i is estimated from external evidence independent of the response data used in Stages 1 and 3. The primary signal is *reference model anchoring*: using a fixed reference model (e.g., gpt-4-0314) as anchor, we compute win-rate residuals for each target model against difficulty-adjusted expectations; items where target models systematically outperform the anchor beyond what difficulty predicts are flagged as potentially exposed. The secondary signal is *cross-model agreement*: items receiving uniformly high scores across architecturally diverse models suggest exposure rather than genuine easiness, quantified via z-score standardization of the inter-model agreement distribution. These signals

are combined as a weighted composite (75% anchoring, 25% agreement), clipped to $[0, 1]$, and treated as fixed in Stage 3. Where model log-probabilities are accessible, LogProber can provide an additional gray-box memorization signal, though it is not used in our current experiments.

Stage 3: Joint MCMC. With e_i fixed, we sample $(\theta_n, b_i, d_j, c_n)$ jointly via NUTS (Betancourt and Girolami, 2015). The priors are as specified in §3.1. For c_n, the regularized horseshoe prior (Piironen and Vehtari, 2017) computes the regularized local scale as

$$\tilde{\lambda}_n^2 = \frac{c_{\text{slab}}^2 \cdot \lambda_n^2}{c_{\text{slab}}^2 + \tau_c^2 \cdot \lambda_n^2}, \quad c_n = |\tau_c \cdot \tilde{\lambda}_n \cdot c_n^{\text{raw}}|, \quad (5)$$

where $c_n^{\text{raw}} \sim \mathcal{N}(0, 1)$. The slab component bounds the maximum effect size, while the Half-Cauchy local scales λ_n provide adaptive shrinkage that concentrates clean models near $c_n = 0$ without attenuating genuinely contaminated ones.

Convergence is assessed via the split-$\hat{R}$ statistic ($\hat{R} < 1.05$) and effective sample size (ESS > 200) across all chains.[1]

3.3 Identifiability

Adding a multiplicative term to a measurement model raises the question of whether the new parameters can be separated from the existing ones. We establish that fixing e_i externally is both necessary and sufficient for identification. All proofs assume the standard location constraint $\sum_n \theta_n = 0$.

Proposition 1 (No New Indeterminacy)**.** *When e_i is fixed externally, the gating term $c_n \cdot e_i$ introduces no additional location or scale indeterminacy beyond that already present in the standard MFRM.*

Proof. Suppose $c_n' = c_n + \delta$ preserves all response probabilities. Then $\delta \cdot e_i = 0$ for all items. Since $e_i \neq 0$ for at least one item by assumption, $\delta = 0$. The key observation is that fixing e_i externally eliminates the bilinear ambiguity $c_n \cdot e_i = (\lambda c_n)(\lambda^{-1} e_i)$ that would otherwise render the decomposition non-unique. □

Proposition 2 (Exposure Variation Necessary)**.** *If $e_i = e$ for all i (constant exposure), then c_n and θ_n are not separately identifiable.*

Proof. When exposure is constant, the linear predictor becomes $(\theta_n + c_n \cdot e) - b_i - d_j - \tau_k$, which is equivalent to a standard MFRM with reparameterized ability $\theta_n^* = \theta_n + c_n \cdot e$. No configuration of response data can distinguish (θ_n, c_n) from $(\theta_n + \epsilon, c_n - \epsilon/e)$ for any ϵ. □

Corollary 1. $\text{Var}(e_i) > 0$ *is a necessary condition for identifiability of c_n.*

Theorem 1 (Sufficient Conditions for Local Identifiability)**.** *The G-MFRM is locally identifiable, up to the standard location constraint, if the following conditions hold: (i) a fully crossed design (all $N \times I \times J$ cells observed), (ii) $J \geq 2$ judges, (iii) at least two distinct exposure values $|\{e_i\}| \geq 2$, (iv) $\text{Var}(e_i) > 0$, and (v) standard normalization $\sum_n \theta_n = 0$.*

Proof sketch. Following the approach of Rothenberg (1971) and San Martín (2016), local identifiability holds if the Fisher information matrix is non-singular at the true parameter values. The critical block is the (θ_n, c_n) submatrix for each model n. Its determinant takes the form

$$\det \mathbf{F}_n = \left(\sum_{i,j} w_{nij}\right)\left(\sum_{i,j} w_{nij} e_i^2\right) - \left(\sum_{i,j} w_{nij} e_i\right)^2, \quad (6)$$

where $w_{nij} > 0$ are the Fisher information weights for each observation. By the Cauchy-Schwarz inequality, this determinant is strictly positive whenever e_i is not constant across items, which is guaranteed by conditions (iii) and (iv). The remaining blocks for b_i and d_j inherit non-singularity from the standard MFRM under conditions (i), (ii), and (v). □

Conditions (i) and (ii) are design requirements satisfied by standard multi-judge evaluation protocols. Conditions (iii) and (iv) are the substantive constraints: the benchmark must contain items with varying levels of exposure. In practice, any benchmark containing both frequently leaked items (e.g., MMLU questions circulated online) and novel items satisfies these conditions.

3.4 Error Propagation

The staged procedure treats the Stage 2 exposure estimates $\hat{e}_i$ as fixed quantities in Stage 3. This is a generated regressors problem (Pagan, 1984): measurement error in $\hat{e}_i$ propagates into the posterior of (θ_n, c_n). The multiplicative structure $c_n \cdot e_i$ makes this an errors-in-variables (EIV) problem (Fuller, 1987), with a predictable attenuation pattern.

[1] Synthetic validation with $N = 50$, $I = 100$, $J = 8$ yields max $\hat{R} = 1.02$ and min ESS $= 276$ across all parameters (exp-012).

Let $\hat{e}_i = e_i + \epsilon_i$ with $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ independent of all model parameters. The attenuation factor for the contamination coefficient is

$$\lambda(\sigma) = \frac{\mathrm{Var}(e_i)}{\mathrm{Var}(e_i) + \sigma^2}, \tag{7}$$

which implies a critical noise threshold $\sigma_{\text{crit}} = \sqrt{\mathrm{Var}(e_i)}$ at which attenuation reaches 50%. In the synthetic DGP of exp-012, $\mathrm{Var}(e_i) \approx 0.04$, yielding $\sigma_{\text{crit}} \approx 0.2$.

Figure **??** and Table **??** show the empirical degradation across noise levels. Below the critical threshold ($\sigma \leq 0.2$), degradation is mild: θ-RMSE increases by 5.9% (from 0.128 ± 0.018 to 0.135 ± 0.018) and c_n-RMSE by 7.9% (from 0.533 ± 0.187 to 0.575 ± 0.198). At $\sigma = 0.1$, the increases are only 1.4% and 2.3% respectively. Beyond the threshold ($\sigma = 0.5$), degradation becomes severe: θ-RMSE rises 36.9% and c_n-RMSE rises 57.9%. Contamination estimates c_n are more sensitive than ability estimates θ_n because c_n enters the likelihood only through its product with e_i (multiplicative coupling), while θ_n contributes additively and is anchored by all observations regardless of exposure.

These bounds make the staged approximation falsifiable: if the noise level of the Stage 2 exposure estimates can be bounded below σ_{crit}, the resulting ability estimates carry quantifiable error guarantees. For honest credible intervals that account for generated-regressor uncertainty, we apply the Murphy-Topel variance correction (Murphy and Topel, 1985) to inflate the Stage 3 posterior variance by the estimated Stage 2 sampling variability.

3.5 Fair Scores and Diagnostics

The Stage 3 posterior mean of θ_n provides a bias-and-contamination-corrected "fair score" for each model. The remaining parameters serve complementary diagnostic roles: per-model c_n flags candidates for training data audit, per-judge d_j informs judge calibration, and per-item e_i identifies items for retirement or replacement.

4 Experiments

5 Related Work

Judge Bias in LLM Evaluation. LLM-as-a-judge protocols introduce systematic scoring differences across judges, and several recent methods target this bias in isolation. CyclicJudge (Zhu et al., 2026) eliminates average bias through round-robin judge assignment that ensures balanced exposure of model pairs to all judges, but treats all residual score variation as genuine ability signal and provides no per-item diagnostics. UDA (Zhang et al., 2026) learns a neural K-factor adjustment that reduces inter-judge standard deviation by 63.4%, yet the learned correction is a black-box transformation with no interpretable decomposition of the severity parameter and no mechanism for contamination effects. Fiedler (2026) formalize judge variability through $J/\Delta J$ diagnostics from classical test theory, identifying sign-reversal failure modes when calibration is shared across judges with heterogeneous severity profiles. These methods model only the judge bias facet: contamination-induced score inflation is absorbed into the estimated abilities, producing the directional bias characterized in Section 2.

Contamination Detection and Adjustment. A parallel line of work quantifies benchmark contamination at the item or model level without accounting for judge effects. ConStat (Dekoninck et al., 2024) compares target model performance against a reference model to estimate contamination effects ($r^2 = 0.94$ with true contamination in controlled settings), establishing a principled quantification framework that G-MFRM's Stage 2 builds upon for exposure estimation. DCR (Xu et al., 2025) adjusts scores on suspected leaked items through fuzzy inference rules, while LogProber (Yax et al., 2024) detects memorization signatures from token-level log-probabilities under gray-box access. Perplexity-based methods such as Min-K% (Shi et al., 2024) offer a simpler detection signal, but ConStat demonstrates that perplexity approaches are fundamentally evadable. All of these methods assume a fixed or averaged rater, so judge severity differences can masquerade as contamination signals. G-MFRM integrates contamination as a gated facet within the measurement model, separating both sources of distortion jointly rather than in sequence.

Preference Leakage. Li et al. (2026) identify a distinct contamination pathway on the judge side: when a judge model evaluates responses from a model in its own family, the judge exhibits inflated preference (same-family preference leakage score $\approx 8.9\%$, same-model $\approx 23.6\%$). This judge-side contamination is complementary to the examinee-side contamination that G-MFRM targets, where the evaluated model has encountered benchmark

items during training. The two phenomena operate at different points in the evaluation pipeline. G-MFRM's judge severity parameter d_j can partially capture preference leakage effects when they manifest as systematic scoring differences across judges, though a full treatment would require judge-model interaction terms beyond the current specification.

IRT and MFRM for LLM Evaluation. Item Response Theory (IRT) provides the psychometric foundation for principled ability estimation in LLM evaluation. Jiao et al. (2025) apply the Many-Facet Rasch Model to compare ten LLM raters with human experts, demonstrating that MFRM separates judge severity from examinee ability in this context, though their model includes no contamination term. PSN-IRT (Zhou et al., 2025) extends IRT to a neural four-parameter logistic framework that scales to large item pools but operates without judge or contamination facets. In educational measurement, Uto and Ueno (2020) show that generalized many-facet models with additional parameters remain identifiable under Hamiltonian Monte Carlo estimation, a result that supports the feasibility of G-MFRM's extended parameterization. The closest methodological antecedent is DG-IRT (Shu et al., 2013), which uses binary gating to detect item preknowledge in educational testing. G-MFRM extends DG-IRT in three respects: continuous rather than binary exposure, explicit judge severity as a nuisance facet, and externally estimated exposure with formal identifiability guarantees (§3.3).

The common gap across these four research threads is the absence of a unified measurement model for judge bias and contamination. G-MFRM bridges this gap through its gated multi-facet structure, eliminating the structural confounding that no sequential pipeline can resolve (§2), while the staged estimation with fixed exposure maintains the identifiability properties of the underlying MFRM.

6 Conclusion

Limitations

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